LDAPA Intelligent Portal — Technical Documentation

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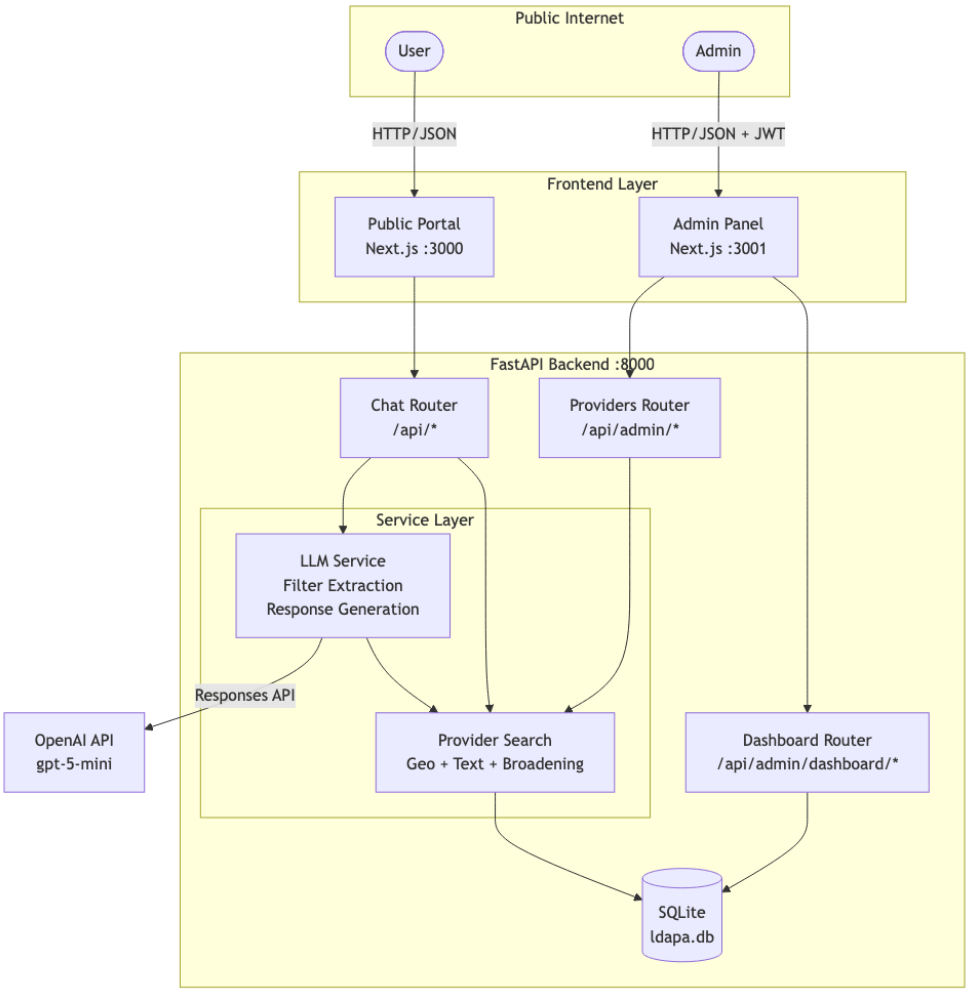
1. System Overview

The LDAPA Intelligent Portal is a conversational AI system for the Learning Disabilities Association of Pennsylvania. It helps parents, adults, educators, and caregivers find learning disability support providers — tutors, health professionals, lawyers, schools, and advocates — through a natural language chat interface.

Core capabilities: - Natural language chat for provider discovery - LLM-powered intent extraction and response generation - Geo-aware provider search with multi-pass broadening - Admin dashboard for provider management and chat analytics - Bulk CSV import from the LDAPA directory export

Current dataset: 3,610 real providers across Pennsylvania (1,529 tutors, 1,138 health professionals, 792 lawyers, 150 schools, 1 advocate).

2. Architecture



System Architecture

Three-application architecture:

Component	Framework	Purpose
Public Portal	Next.js 16	Chat interface for end users
Admin Panel	Next.js 16	Provider management, chat review, analytics
Backend API	FastAPI	REST API, LLM orchestration, database

All frontend-backend communication is via JSON over HTTP. The backend is the single source of truth and the only component that talks to the database and OpenAI.

3. Tech Stack

Backend

Dependency	Version	Purpose
Python	3.10+	Runtime
FastAPI	0.115.0	Web framework, async request handling
Uvicorn	0.30.0	ASGI server
aiosqlite	0.20.0	Async SQLite driver
OpenAI SDK	2.22.0	LLM API client (Responses API)
Pydantic	2.9.0	Request/response validation
python-jose	3.3.0	JWT token encoding/decoding
bcrypt	4.2.0	Password hashing
python-dotenv	1.0.1	Environment variable loading
python-multipart	0.0.9	File upload parsing

Frontend (both apps)

Dependency	Version	Purpose
Next.js	16.1.6	React framework with SSR/SSG
React	19.2.3	UI library
Tailwind CSS	4.x	Utility-first CSS
TypeScript	5.x	Type safety

Database

Technology	Purpose
SQLite	Primary database (dev & production for moderate scale)
WAL mode	Concurrent read access

External Services

Service	Purpose
OpenAI API	Filter extraction + response generation (gpt-5-mini)

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4. Data Schema

4.1 Providers Table

The providers table is the core of the system. It stores all LDAPA directory members.

```sql
CREATE TABLE providers (
    -- Identity
    id                  TEXT PRIMARY KEY,      -- UUID, generated on import/create
    first_name          TEXT,                  -- From CSV first_name
    last_name           TEXT,                  -- From CSV last_name
    name                TEXT NOT NULL,         -- Computed: "{first_name}
        {last_name}"
    listing_type        TEXT,                  -- "Individual" or "Company"

    -- Classification
    profession_name     TEXT NOT NULL,         -- Primary category (see below)
    services            TEXT,                  -- Sub-services, e.g.
        "Therapist,Doctor=>Psychiatrist"
    training            TEXT,                  -- Methodology: "Wilson Language",
        "Orton-Gillingham"
    credentials         TEXT,                  -- License numbers: "Pennsylvania /
        SP023530"
    license             TEXT,                  -- License type: "Psychologist"

    -- Location
    address             TEXT,                  -- Raw address from CSV address1
    city                TEXT,                  -- Parsed from address1 (34% fill
        rate)
    state               TEXT DEFAULT 'PA',     -- Full state name
    state_code          TEXT DEFAULT 'PA',     -- 2-letter code
    zip_code            TEXT,                  -- 5-digit ZIP (90% fill rate)
    lat                 REAL,                 -- Latitude (88% fill rate)
    lon                 REAL,                 -- Longitude (88% fill rate)

    -- Ages & Grades
    age_range_served    TEXT,                  -- Free text: "Adults, Teens" or "4-
        65"
    grades_offered      TEXT,                  -- School-only: "Grades 1-8"

    -- Pricing & Insurance
    price_per_visit     TEXT,                  -- Range: "$150 - $250" (22% fill
        rate)
    sliding_scale       INTEGER DEFAULT 0,     -- Boolean flag (14% fill rate)
    insurance_accepted TEXT,                   -- Free text: "Aetna, Cigna,
        Highmark" (21% fill rate)

    -- Specialty Flags
    ld_adhd_specialty            INTEGER DEFAULT 0,  -- LD/ADHD specialist flag
    learning_difference_support  INTEGER DEFAULT 0,  -- Supports learning
        differences
    adhd_support                 INTEGER DEFAULT 0,  -- Supports ADHD

    -- School-Specific (NULL for non-schools)
    student_body_type TEXT,                    -- "Co-Ed", "All-boys", "All-girls"
    total_size          TEXT,                  -- "25 students"
    average_class_size TEXT,
    religion            TEXT,                  -- "Nonsectarian"
```

```sql
    -- Contact
    phone             TEXT,                 -- Phone number (38% fill rate)
    email             TEXT,                 -- Email, filtered for placeholders
        (89% fill rate)
    website           TEXT,                 -- Website URL (39% fill rate)
    profile_url       TEXT,                 -- LDAPA directory profile link

    -- Admin
    verification_status TEXT NOT NULL DEFAULT 'unverified',  --
        verified/unverified/archived
    last_verified_at    TEXT,
    staff_notes         TEXT,
    created_at          TEXT NOT NULL DEFAULT (datetime('now')),
    updated_at          TEXT NOT NULL DEFAULT (datetime('now')),
    is_deleted          INTEGER NOT NULL DEFAULT 0           -- Soft delete flag
);
```

Indexes: - `idx_providers_city` — city text match - `idx_providers_zip` — ZIP code lookup - `idx_providers_state` — state filtering - `idx_providers_status` — verification status - `idx_providers_profession` — profession type filtering

Profession Name Values

Value	Count	Description
`Tutor`	1,529	Reading, math, learning support tutors
`Health_Professional`	1,138	Therapists, psychologists, psychiatrists, counselors
`Lawyer`	792	Education/disability law attorneys
`School`	150	Private schools and specialized programs
`Advocate`	1	Educational advocates

Where Each Field Is Used

Field	Chat Search	Provider Cards	Admin List	Admin Detail	LLM Context	CSV Impor
`name`	relevance ranking	display	display	display	yes	mappe
`profession_name`	WHERE filter	badge	type column	display	yes	direct
`services`	—	expanded detail	—	display	yes	direct
`training`	LIKE filter	expanded detail	—	display	yes	direct
`credentials`	—	expanded detail	—	display	yes	direct
`city`	WHERE filter	subtitle	location column	display	yes	parse
`state_code`	fallback filter	subtitle	location column	display	yes	direct
`zip_code`	geo lookup	subtitle	—	display	yes	direct
`lat` / `lon`	proximity search	—	—	—	—	direct
`age_range_served`	LIKE filter	expanded detail	—	display	yes	direct
`price_per_visit`	—	subtitle	—	display	yes	direct
`sliding_scale`	WHERE filter	badge	—	display	yes	parse
`insurance_accepted`	LIKE filter	expanded detail	—	display	yes	direct
`ld_adhd_specialty`	WHERE filter	—	—	display	—	parse
`learning_difference_support`	WHERE filter	—	—	display	—	parse
`adhd_support`	WHERE filter	—	—	display	—	parse

Field	Chat Search	Provider Cards	Admin List	Admin Detail	LLM Context	CSV Impor
`grades_offered`	—	expanded detail	—	display	yes	direct
`phone`	—	contact link	—	display	yes	direct
`email`	—	contact link	—	display	—	mappe
`website`	—	contact link	—	display	yes	direct
`verification_status`	WHERE filter	—	status badge	display	—	defaul

4.2 Chat Sessions Table

Tracks anonymous chat conversations.

```
CREATE TABLE chat_sessions (
    id              TEXT PRIMARY KEY,      -- UUID
    user_location   TEXT,                  -- JSON:
        {"city":"Pittsburgh","zip":"15213"}
    started_at      TEXT NOT NULL,
    last_message_at TEXT NOT NULL,
    message_count   INTEGER NOT NULL DEFAULT 0,
    escalated       INTEGER NOT NULL DEFAULT 0  -- Crisis flag
);
```

Used by: Chat router (create/update), Dashboard (stats, recent sessions, session detail).

4.3 Chat Messages Table

Stores every message in every conversation.

```
CREATE TABLE chat_messages (
    id              TEXT PRIMARY KEY,
    session_id      TEXT NOT NULL REFERENCES chat_sessions(id),
    role            TEXT NOT NULL,          -- "user" or "assistant"
    content         TEXT NOT NULL,
    providers_shown TEXT DEFAULT '[]',      -- JSON array of provider IDs
    created_at      TEXT NOT NULL
);
```

Used by: Chat router (store messages), Dashboard (session detail, theme extraction).

4.4 Chat Feedback Table

Stores thumbs-up/down feedback on assistant messages.

```
CREATE TABLE chat_feedback (
    id          TEXT PRIMARY KEY,
    message_id  TEXT NOT NULL REFERENCES chat_messages(id),
    session_id  TEXT NOT NULL REFERENCES chat_sessions(id),
    rating      TEXT NOT NULL,            -- "up" or "down"  (note: API accepts
        "positive"/"negative")
    created_at  TEXT NOT NULL
);
```

Used by: Chat feedback endpoint, Dashboard (average rating calculation).

4.5 Admin Users Table

```
CREATE TABLE admin_users (
    id            TEXT PRIMARY KEY,
    email         TEXT NOT NULL UNIQUE,
    password_hash TEXT NOT NULL,          -- bcrypt hash
    name          TEXT NOT NULL,
    created_at    TEXT NOT NULL
);
```

Default credentials: `admin@ldapa.org` / `admin123` (change in production).

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5. Agentic Flow — Chat Pipeline

5.1 Why Two LLM Calls?

Every user message triggers a pipeline that makes **two separate LLM calls** with a database query in between. This separation exists because the system needs to do two fundamentally different things:

1. **Understand what the user wants** (structured data extraction) — the LLM reads the full conversation and outputs a JSON object describing the user's intent: what type of provider they need, where they are, what age group, whether they mentioned insurance, etc. This is a classification task that benefits from constrained JSON output.

2. **Respond naturally** (free-form text generation) — the LLM takes the conversation history *plus the actual provider results from the database* and writes a helpful, empathetic response. It can only do this after the database query runs, because it needs to know which providers were found (or not found) to craft an appropriate answer.

The database query sits between these two calls — it uses the structured filters from Call 1 to search for matching providers, which are then fed into Call 2 as context.

5.2 How a Single Chat Message Is Processed

When a user sends a message, here is exactly what happens:

Step 1 — Rate limiting and session setup. The system checks that the user hasn't exceeded 30 requests per minute. If this is the first message, a new session is created with a UUID. The session tracks message count, user location, and whether the conversation has been escalated.

Step 2 — Build conversation context. The full conversation history (all prior user and assistant messages) is assembled into an array, with the new message appended. This entire history is passed to both LLM calls so the model has full context — it never sees just the latest message in isolation.

Step 3 — LLM Call 1: Filter Extraction. The conversation history is sent to the OpenAI Responses API (gpt-5-mini, reasoning effort: low, JSON output mode) along with the `FILTER_EXTRACTION_PROMPT`. This prompt instructs the model to analyze the conversation and output a structured JSON object containing: - What type of provider the user wants (`profession_types`) - Any specializations mentioned (`specializations`) - Location information (`city`, `zip`) - Age group of the person needing help - Whether a specific insurance provider was mentioned - Whether a specific training methodology was requested (e.g., Orton-Gillingham) - Three routing flags: `needs_providers`, `needs_more_info`, `escalate`

The raw JSON output is then validated — values are checked against allowed lists, types are coerced, and invalid entries are silently dropped.

Step 4 — Routing decision. Based on the extracted filters, the system takes one of three paths:

- **Escalation** (`escalate = true`): The user mentioned crisis, self-harm, abuse, or something that requires human intervention. The session is flagged as escalated, and the response will direct them to LDAPA staff and crisis resources. No provider search happens.

- **Needs more info** (`needs_more_info = true` and fewer than 3 user turns): The user seems to want a provider but hasn't given enough information to search effectively — no location, no service type, no age group. The response will ask 1-2 follow-up questions. No provider search happens. *However*, after 3 user turns, this flag is overridden and a search is forced anyway, to prevent the system from endlessly asking questions.

- **Provider search** (`needs_providers = true`): The user is actively looking for a provider and has given enough information. The system queries the database.

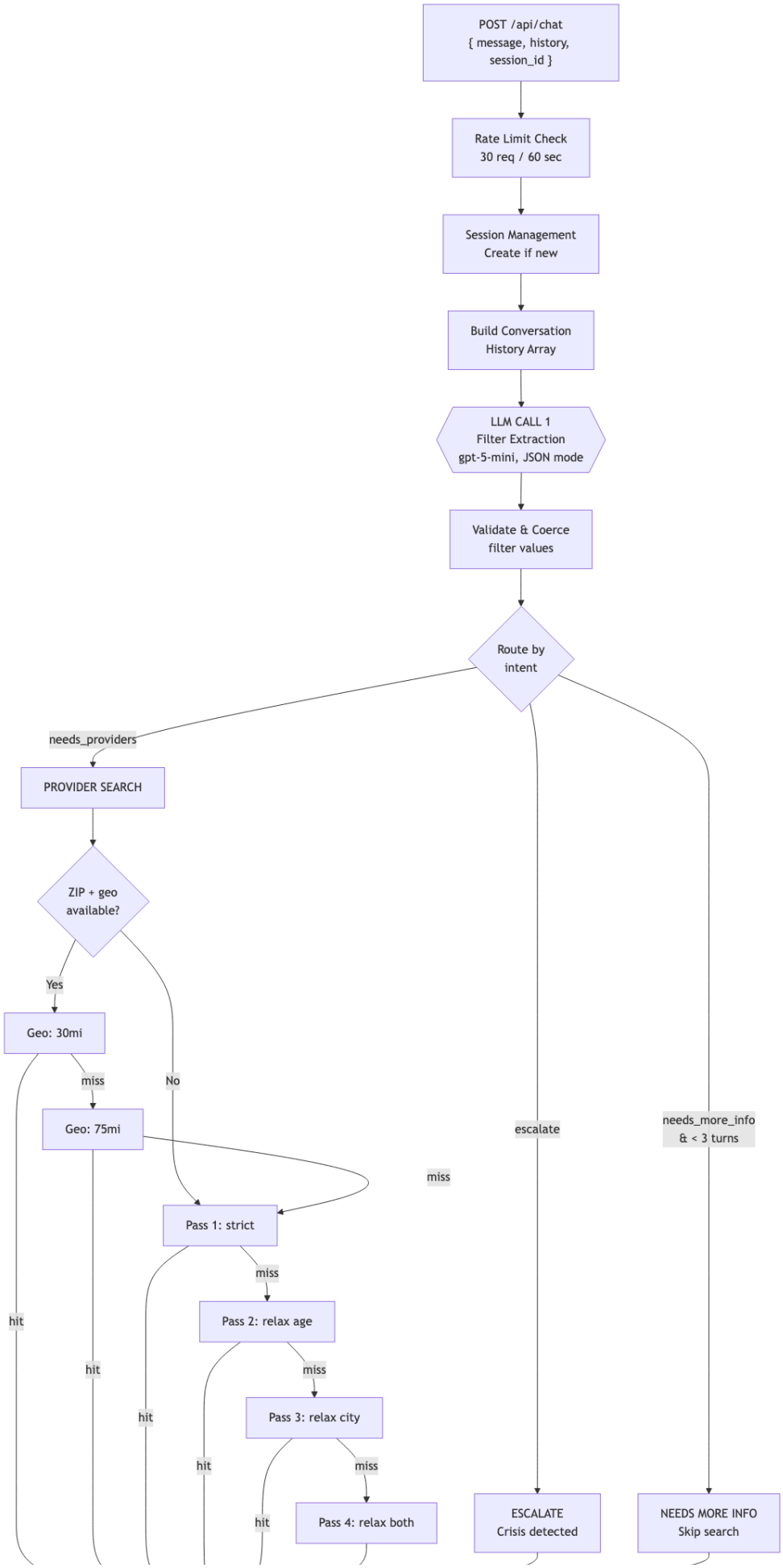
Step 5 — Provider search (if triggered). The search engine runs a multi-pass strategy to find up to 5 matching providers. It first attempts geo-based proximity search if a ZIP code was provided and lat/lon data is available. If that yields no results (or no ZIP was given), it falls through to text-based matching with progressive filter relaxation — first strict (exact city + age), then relaxing age, then relaxing city to state-wide, then relaxing both. Lawyers are deprioritized in results unless the user specifically asked for legal help. Full details in Section 7.

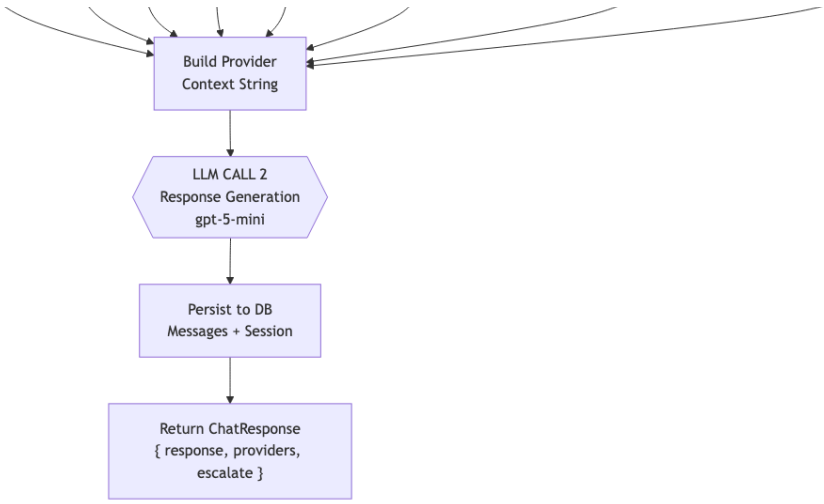
Step 6 — Build provider context string. The search results (or the reason no search was done) are formatted into a plain text string that will be injected into the response generation prompt. This string tells the LLM exactly what happened: - `"_ESCALATE_"` — crisis detected - `"Need more information from the user..."` — not enough info to search - `"No provider search needed..."` — user asked a general question - `"No matching providers found..."` — search ran but found nothing - A formatted list of providers with their details (name, type, training, location, pricing, insurance, phone, website)

Step 7 — LLM Call 2: Response Generation. The conversation history is sent again to the OpenAI Responses API, this time with the `RESPONSE_GENERATION_PROMPT` and the provider context string interpolated into it. This prompt defines the assistant's persona (friendly LDAPA guide), its boundaries (never diagnose, never provide legal advice), and five response patterns based on the context type. The model generates a natural language response. If providers were found, the response includes a `[PROVIDERS]` marker that the frontend strips out and replaces with rendered provider cards.

Step 8 — Persist and respond. Both the user message and the assistant response are saved to the `chat_messages` table with the session ID. The provider IDs shown are stored as a JSON array. The session's message count and last-activity timestamp are updated. If location was extracted, it's saved to the session. Finally, the `ChatResponse` is returned containing the response text, an array of provider card objects (for frontend rendering), and the escalation flag.

5.3 Flow Diagram


POST /api/chat
{ message, history,
session_id }
Rate Limit Check
30 req / 60 sec
Session Management
Create if new
Build Conversation
History Array
LLM CALL 1
Filter Extraction
gpt-5-mini, JSON mode
Validate & Coerce
filter values
Route by
intent
needs_providers
PROVIDER SEARCH
ZIP + geo
available?
Yes
No
Geo: 30mi
miss
Geo: 75mi
hit
hit
miss
Pass 1: strict
hit
miss
Pass 2: relax age
hit
miss
Pass 3: relax city
hit
miss
Pass 4: relax both
hit
escalate
ESCALATE
Crisis detected
needs_more_info
& < 3 turns
NEEDS MORE INFO
Skip search



Chat Pipeline

5.4 Fallback Mode (No OpenAI Key)

When `OPENAI_API_KEY` is not set, the system uses keyword-based fallbacks so the application remains functional without LLM access:

- **Filter extraction:** Regex and keyword matching against hardcoded lists of profession types, specializations, PA city names, ZIP patterns, insurance provider names, and training methodologies. Less accurate than the LLM but covers common cases.
- **Response generation:** Template-based responses with four paths: escalation (directs to crisis lines), providers found (generic intro + `[PROVIDERS]` marker), no providers (suggests contacting LDAPA directly), and general question (asks follow-up questions).

5.5 Auto-Search After 3 Turns

After 3 or more user messages, the system overrides `needs_more_info` and forces a provider search with whatever filters have been gathered so far. This prevents the system from endlessly asking clarifying questions when the user may not know exactly what they need — it's better to show imperfect results than to keep asking.

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6. LLM Prompts & Filter Schema

6.1 Filter Extraction Prompt

Sent as the system/developer message in LLM Call 1. Instructs the model to return structured JSON.

Output schema:

```
{
  "profession_types": [],
  "specializations": [],
  "insurance": null,
  "training_methodology": null,
  "sliding_scale": false,
  "location": {
    "city": null,
    "zip": null
  },
  "age_group": [],
  "needs_providers": false,
  "needs_more_info": false,
  "escalate": false,
  "search_text": ""
}
```

Allowed values:

Field	Values
`profession_types`	`tutor`, `health_professional`, `lawyer`, `school`, `advocate`
`specializations`	`dyslexia`, `adhd`, `ld`, `learning_differences`
`insurance`	Any insurer name (free text)
`training_methodology`	Any methodology name (free text)
`age_group`	`children` (<13), `adolescents` (13-17), `adults` (18+)

Key prompt rules: - Extract from entire conversation history, not just last message - Map user language to profession types (e.g., “therapist” -> `health_professional`) - `needs_providers=true` only for active provider-seeking intent - `needs_more_info=true` only when no useful filters can be extracted at all - `escalate=true` for crisis, self-harm, abuse - Never re-ask for information already provided

6.2 Response Generation Prompt

Sent as the system/developer message in LLM Call 2. Includes the provider context string interpolated into `{provider_context}`.

Five response modes based on context:

1. **General Question** — Answer directly, no provider search
2. **Missing Information** — Ask 1-2 follow-up questions conversationally

3. **Providers Found** — Present results with `[PROVIDERS]` marker for card rendering
4. **No Providers Found** — Honest about limitation, suggest LDAPA contact
5. **Escalation** — Empathetic redirect to crisis services (988, 911)

Boundaries enforced in prompt: - Never diagnose, provide legal/medical advice - Never suggest searching external sites - Never claim to be a licensed professional - Always include disclaimer on first response

6.3 Filter Validation

After LLM output is parsed as JSON, `_validate_filters()` coerces values: - Arrays are filtered against allowed value sets - Strings are trimmed and limited to 200 chars - Booleans are cast with `bool()` - Location is validated as a dict with `city` / `zip` keys

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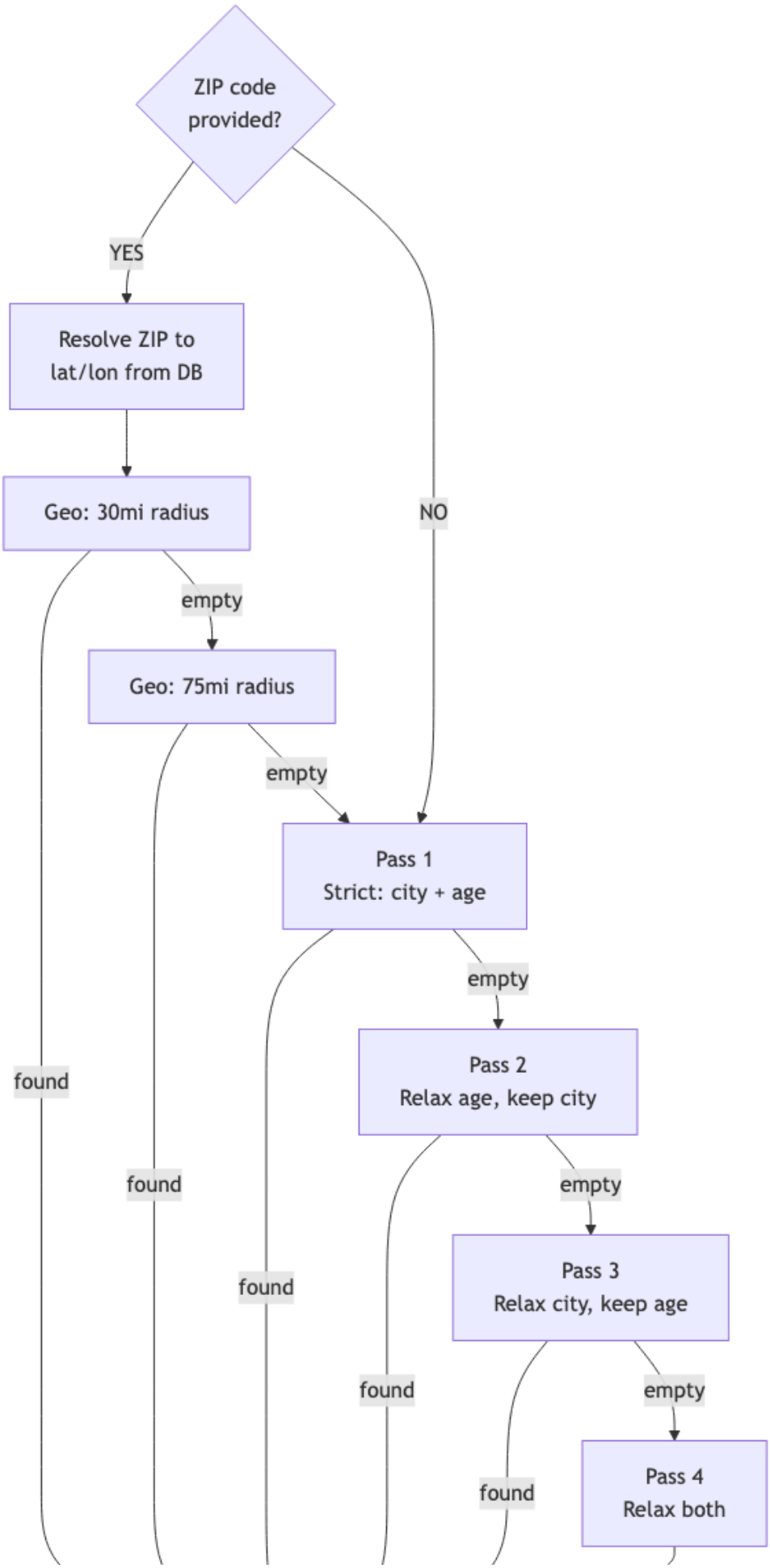
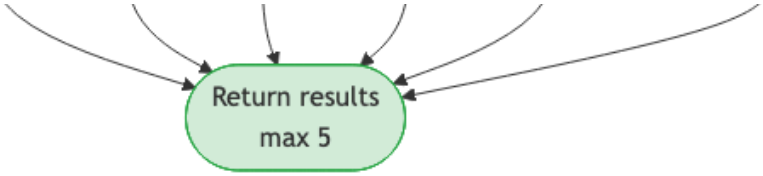
7. Provider Search Engine

7.1 Search Strategy

The search engine uses a **tiered fallback** approach to maximize result relevance while ensuring users always get results.


ZIP code provided?
YES
NO
Resolve ZIP to lat/lon from DB
Geo: 30mi radius
empty
found
Geo: 75mi radius
empty
found
Pass 1
Strict: city + age
empty
found
Pass 2
Relax age, keep city
empty
found
Pass 3
Relax city, keep age
empty
found
Pass 4
Relax both



Search Strategy

7.2 Geo Search

Uses approximate planar distance (valid for PA-scale distances):

- `MILES_PER_DEG_LAT = 69.0`
- `MILES_PER_DEG_LON = 53.0` (cos(40°) × 69)
- Bounding box pre-filter for performance
- Squared distance for ranking (avoids sqrt)

ZIP-to-coordinates resolution uses `AVG(lat), AVG(lon)` from existing providers with that ZIP.

7.3 Lawyer Deprioritization

Unless the user explicitly asks for legal help (detected by `profession_types` containing `lawyer`), lawyers are sorted after other provider types using:

```
ORDER BY CASE WHEN p.profession_name = 'Lawyer' THEN 1 ELSE 0 END, ...
```

7.4 Filter Mapping to SQL

Filter	SQL Condition
`profession_types`	`LOWER(profession_name) IN (?)`
`specializations: adhd`	`adhd_support = 1 OR ld_adhd_specialty = 1`
`specializations: dyslexia/ld`	`learning_difference_support = 1 OR ld_adhd_specialty = 1` + `LIKE` on training
`training_methodology`	`LOWER(training) LIKE ?`
`insurance`	`LOWER(insurance_accepted) LIKE ?`
`sliding_scale`	`sliding_scale = 1`
`age_group: children`	`LOWER(age_range_served) LIKE '%child%' OR '%preteens%' OR '%k-%'`
`age_group: adolescents`	`LOWER(age_range_served) LIKE '%teen%' OR '%adolesc%'`
`age_group: adults`	`LOWER(age_range_served) LIKE '%adult%' OR '%elder%'`
`location.city`	`LOWER(city) = LOWER(?)`
`location.zip`	`zip_code = ?` (or geo search if lat/lon available)

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8. Authentication & Authorization

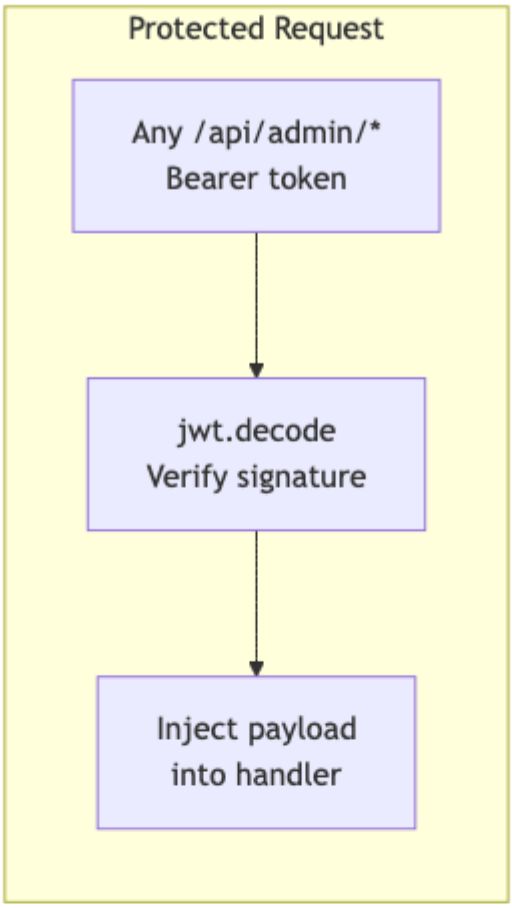
Public Routes (no auth)

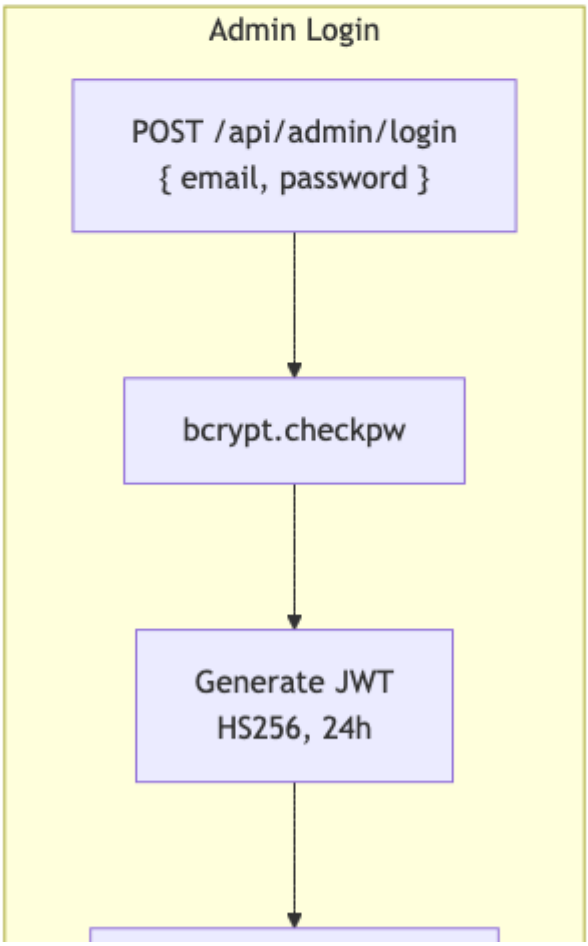
- `POST /api/chat` — Chat endpoint (rate-limited: 30/60s)
- `POST /api/feedback` — Feedback submission
- `GET /api/health` — Health check

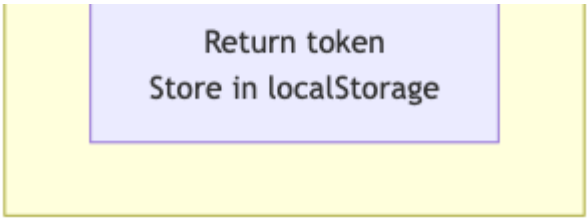
Admin Routes (JWT required)

All `/api/admin/*` routes require a `Bearer` token in the `Authorization` header.

Auth flow:


Protected Request
Any /api/admin/*
Bearer token
jwt.decode
Verify signature
Inject payload
into handler


Admin Login
POST /api/admin/login
{ email, password }
bcrypt.checkpw
Generate JWT
HS256, 24h



Auth Flow

Client-side (Admin Panel): - Token stored in `localStorage` as `ldapa_admin_token` - User object stored as `ldapa_admin_user` - `authFetch()` wrapper auto-attaches Bearer token - 401 responses redirect to `/login`

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9. Admin Dashboard

Dashboard Page (`/dashboard`)

Displays four stat cards and two panels: - **Total Providers** (with verified count) - **Unverified** count - **Chat Sessions** (filtered by period: week/month/all) - **Average Feedback** (0-5 scale, computed from up/down ratio) - **Chat Volume** — Bar chart of sessions per day - **Top Query Themes** — Keyword-extracted from recent user messages

Providers Management (`/dashboard/providers`)

- Table with name, type, location, status, last verified
- Search by name/services/training
- Filter by status (verified/unverified/archived)
- Filter by profession type
- Bulk verify/archive with multi-select
- Individual provider detail page with all fields
- Edit form with all fields (school-specific fields shown conditionally)
- Soft delete

Chat Review (`/dashboard/chats`)

- List of all sessions with message count, location, escalation flag, avg rating
- Click to view full session transcript with messages and feedback

CSV Import (`/dashboard/providers/import`)

- Three-step wizard: Upload → Preview → Done
- Shows validation errors and warnings
- Preview table of first 50 providers
- Confirm to bulk insert

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10. CSV Data Import Pipeline

Source Format

The LDAPA directory export CSV contains 130+ columns. The importer maps the relevant subset:

CSV Column	DB Field	Transformation
`first_name` + `last_name`	`name`	Concatenated with space
`profession_name`	`profession_name`	Direct (validated against 5 values)
`listing_type`	`listing_type`	Direct
`address1`	`address` , `city`	Address stored as-is; city parsed via regex
`state_ln`	`state`	Direct, default “Pennsylvania”
`state_code`	`state_code`	Direct, default “PA”
`zip_code`	`zip_code`	Direct
`lat` , `lon`	`lat` , `lon`	Parsed as float, 0 → NULL
`services`	`services`	Direct
`training`	`training`	Direct
`credentials` / `Credentials`	`credentials`	First non-empty wins
`license`	`license`	Direct
`agerangeserved`	`age_range_served`	Direct
`pricepervisit`	`price_per_visit`	Direct
`Slidingscaleoffered`	`sliding_scale`	Parsed as bool (handles “0”, “1”, “$1.00”)
`insurancesaccepted`	`insurance_accepted`	Direct
`LDADHDspeciality`	`ld_adhd_specialty`	Parsed as bool
`learning_difference_support`	`learning_difference_support`	Parsed as bool
`adhd_support`	`adhd_support`	Parsed as bool
`Email` / `email`	`email`	First non-empty; filters `@ldaofpadirectory.com`
`phone_number` / `PhoneNumber`	`phone`	First non-empty
`website` / `Website`	`website`	First non-empty

City Parsing

The `city` column in the CSV is nearly empty (2 rows). Cities are parsed from `address1` with regex:

1. Match `City, PA` pattern (e.g., “123 Main St, Pittsburgh, PA 15213”)
2. Match standalone city name (e.g., “East Stroudsburg”)
3. Fallback to `city` column, then `City` column
4. NULL if no match (66% of rows)

Fill rates after import: - City parsed: 34% (1,240/3,610) - Lat/lon available: 88% (3,166/3,610) — compensates for missing cities via geo search

Auto-Import on Startup

`init_db()` checks if the providers table is empty. If so, it searches the project root for a CSV file matching `*export_members*.csv` and imports it automatically.

---

11. API Reference

Public Endpoints

Method	Path	Description
`POST`	`/api/chat`	Send chat message, receive response + providers
`POST`	`/api/feedback`	Submit thumbs up/down on a message
`GET`	`/api/health`	Health check

Admin Endpoints (require JWT)

Method	Path	Description
`POST`	`/api/admin/login`	Authenticate, receive JWT
`GET`	`/api/admin/dashboard/stats`	Dashboard statistics
`GET`	`/api/admin/dashboard/chat-volume`	Chat volume time series
`GET`	`/api/admin/dashboard/recent-sessions`	Paginated session list
`GET`	`/api/admin/dashboard/sessions/{id}`	Full session detail
`GET`	`/api/admin/providers`	List providers (paginated, filterable)
`GET`	`/api/admin/providers/{id}`	Provider detail
`POST`	`/api/admin/providers`	Create provider
`PUT`	`/api/admin/providers/{id}`	Update provider
`DELETE`	`/api/admin/providers/{id}`	Soft-delete provider
`PATCH`	`/api/admin/providers/{id}/verify`	Mark verified
`PATCH`	`/api/admin/providers/{id}/archive`	Mark archived
`POST`	`/api/admin/providers/bulk-verify`	Bulk verify
`POST`	`/api/admin/providers/bulk-archive`	Bulk archive
`POST`	`/api/admin/providers/import/preview`	Preview CSV import
`POST`	`/api/admin/providers/import/confirm`	Confirm CSV import

---

12. Environment Configuration

Backend (`backend/.env`)

```
DATABASE_PATH=ldapa.db                    # SQLite file path
OPENAI_API_KEY=sk-...                     # Required for LLM features
JWT_SECRET=<random-64-char-string>        # Required in production
CORS_ORIGINS=https://portal.ldapa.org,https://admin.ldapa.org
LLM_MODEL=gpt-5-mini                     # OpenAI model ID
```

Public Portal (`public-portal/.env`)

```
NEXT_PUBLIC_API_URL=https://api.ldapa.org  # Backend URL
```

Admin Panel (`admin-panel/.env`)

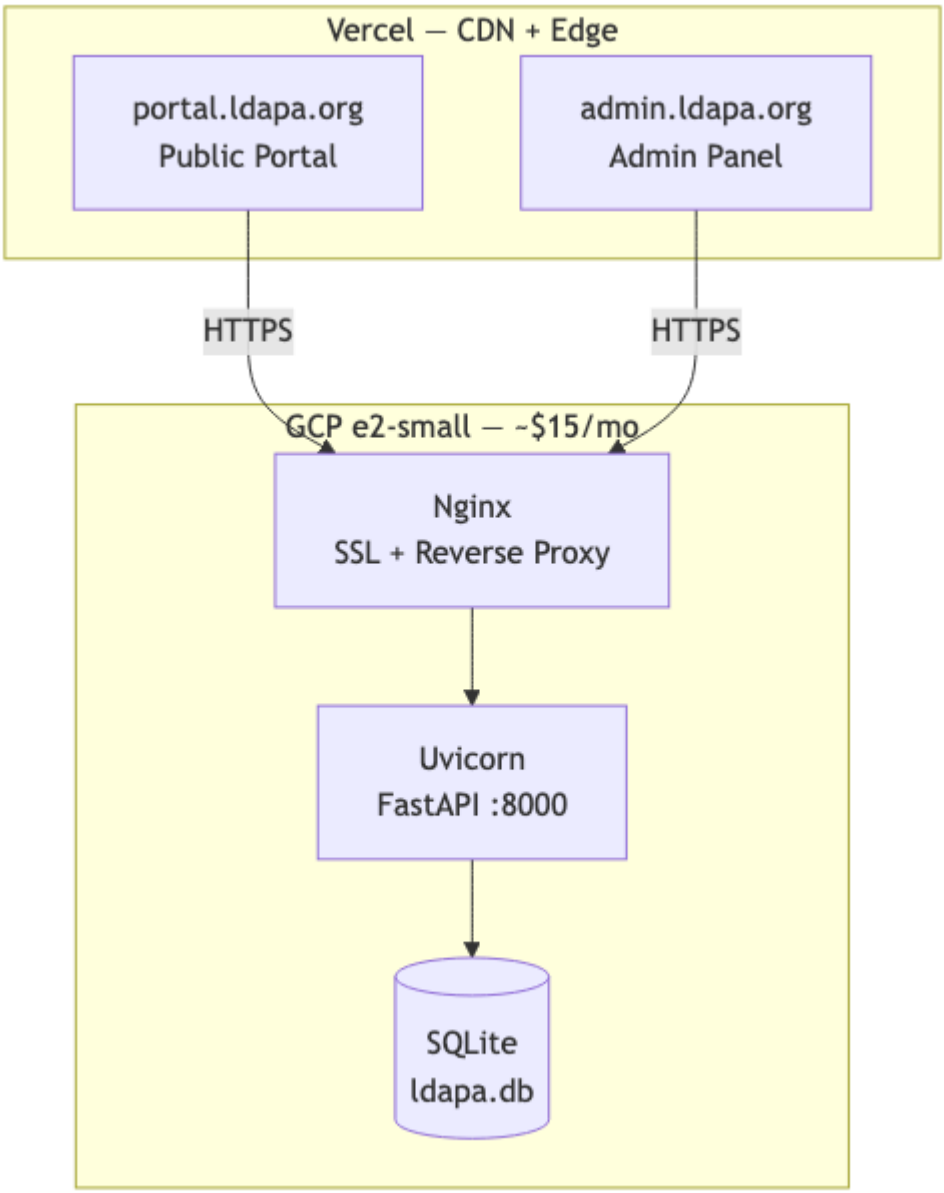
```
NEXT_PUBLIC_API_URL=https://api.ldapa.org  # Backend URL
```

---

13. Deployment

Option A: Vercel (Frontends) + GCP VM (Backend)

Best for: Leveraging Vercel's CDN and zero-config deployments for the Next.js apps, while keeping the backend on a VM for full control over the database and LLM calls.



Deployment Option A

GCP VM Setup:

```
# Recommended: e2-small (2 vCPU, 2 GB RAM) — ~$15/month
# OS: Ubuntu 22.04 LTS

# Install dependencies
sudo apt update && sudo apt install -y python3.11 python3.11-venv nginx certbot

# Clone and setup
git clone <repo> /opt/ldapa-portal
cd /opt/ldapa-portal/backend
python3.11 -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# Create systemd service
sudo cat > /etc/systemd/system/ldapa.service << 'EOF'
[Unit]
Description=LDAPA Backend API
After=network.target

[Service]
User=ldapa
WorkingDirectory=/opt/ldapa-portal/backend
Environment=PATH=/opt/ldapa-portal/backend/venv/bin
EnvironmentFile=/opt/ldapa-portal/backend/.env
ExecStart=/opt/ldapa-portal/backend/venv/bin/uvicorn app.main:app --host
        127.0.0.1 --port 8000 --workers 2
Restart=always

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl enable ldapa && sudo systemctl start ldapa
```

Nginx config (`/etc/nginx/sites-available/ldapa`):

```
server {
    listen 443 ssl http2;
    server_name api.ldapa.org;

    ssl_certificate /etc/letsencrypt/live/api.ldapa.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/api.ldapa.org/privkey.pem;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

Vercel deployment:

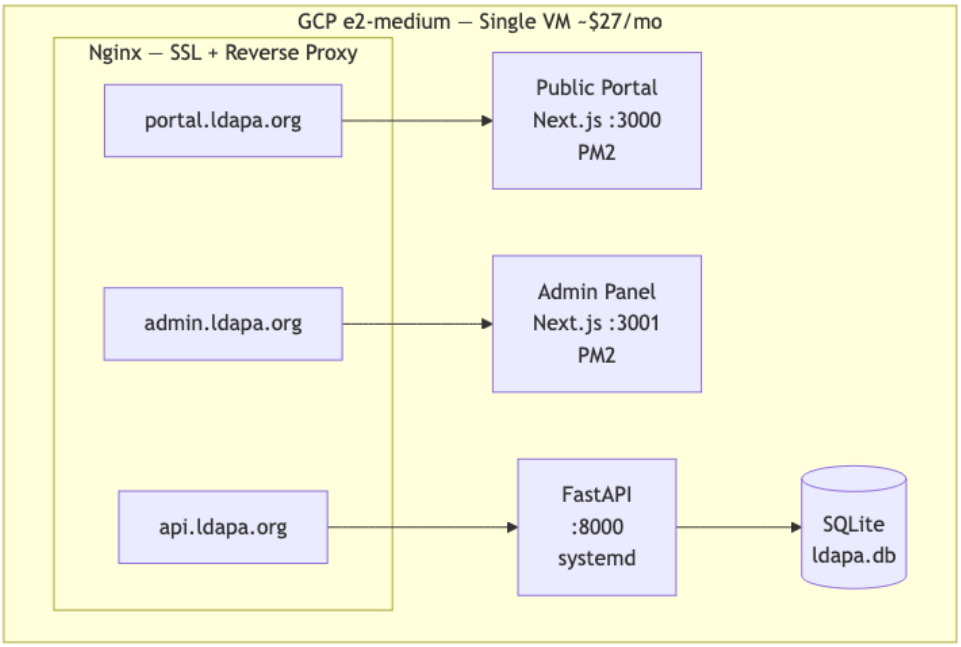
```
# Public portal
cd public-portal
vercel --prod
# Set env var: NEXT_PUBLIC_API_URL=https://api.ldapa.org

# Admin panel
cd admin-panel
vercel --prod
# Set env var: NEXT_PUBLIC_API_URL=https://api.ldapa.org
```

Pros: - Vercel handles CDN, SSL, and scaling for frontends automatically - Global edge distribution for static assets - Backend has full control: file system, SQLite, long-running processes

Cons: - Two platforms to manage - Cross-origin setup required (CORS already handled) - Backend SSL must be managed manually (Certbot)

Option B: Single VM (Everything)

Best for: Simpler operations. One machine, one set of logs, one place to SSH into.



Deployment Option B

VM specs: e2-medium (2 vCPU, 4 GB RAM) — ~$27/month

Setup:

```
# Install Node.js, Python, Nginx
sudo apt update
sudo apt install -y python3.11 python3.11-venv nginx certbot
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
sudo npm install -g pm2

# Build frontends
cd /opt/ldapa-portal/public-portal
npm ci && npm run build
cd /opt/ldapa-portal/admin-panel
npm ci && npm run build

# Start frontends with PM2
pm2 start npm --name "public-portal" -- start -- --port 3000
pm2 start npm --name "admin-panel" -- start -- --port 3001
pm2 save && pm2 startup

# Backend (same as Option A systemd setup)
```

Nginx config for all three services:

```
# Public portal
server {
    listen 443 ssl http2;
    server_name portal.ldapa.org;
    ssl_certificate /etc/letsencrypt/live/portal.ldapa.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/portal.ldapa.org/privkey.pem;

    location / {
        proxy_pass http://127.0.0.1:3000;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

# Admin panel
server {
    listen 443 ssl http2;
    server_name admin.ldapa.org;
    ssl_certificate /etc/letsencrypt/live/admin.ldapa.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/admin.ldapa.org/privkey.pem;

    location / {
        proxy_pass http://127.0.0.1:3001;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

# API
server {
    listen 443 ssl http2;
    server_name api.ldapa.org;
    ssl_certificate /etc/letsencrypt/live/api.ldapa.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/api.ldapa.org/privkey.pem;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

Pros: - Single machine to manage, monitor, and back up - No cross-origin complexity if using same domain - Lower total cost - SQLite backup is just copying a file

Cons: - No CDN for frontend assets (can add Cloudflare in front) - Single point of failure - Must manage Node.js + Python on same machine

---

Deployment Comparison

Aspect	Option A (Vercel + GCP)	Option B (Single VM)
Monthly cost	~$15 VM + $0 Vercel (hobby)	~$27 VM
SSL	Automatic (Vercel) + Certbot (VM)	Certbot for all three
CDN	Vercel Edge (global)	None (add Cloudflare)
Scaling frontends	Automatic	Manual (upgrade VM)
Operational complexity	Two platforms	One machine
Deployment	`vercel --prod` + SSH	SSH only
Backup	VM snapshot + SQLite copy	VM snapshot
Monitoring	Vercel dashboard + VM metrics	VM metrics only

Recommendation for moderate traction: Option B (single VM) with Cloudflare in front for CDN/DDoS. Simpler to operate and debug. Switch to Option A if frontend traffic grows significantly.

---

14. Scaling Considerations

Current Capacity (SQLite, single VM)

Metric	Capacity
Providers	Tested with 3,610; SQLite handles millions
Concurrent chat users	~50-100 (limited by LLM API rate)
Requests/sec	~200 (FastAPI + aiosqlite)
Database size	~5 MB for 3,610 providers
LLM calls per chat message	2 (filter extraction + response)

Moderate Growth (5,000-20,000 providers, 500 daily chats)

No architecture changes needed. SQLite in WAL mode handles this easily.

Change	When	Effort
Add Cloudflare CDN	Immediately	Low — DNS change
Upgrade VM to e2-medium (4 GB)	>100 concurrent users	Low — resize
Add SQLite FTS5 for full-text search	>10,000 providers	Medium — schema change
Add response caching (Redis)	>1,000 daily chats	Medium

Significant Growth (50,000+ providers, 5,000+ daily chats)

Change	Trigger	Effort
Migrate SQLite → PostgreSQL	Write contention or >100K rows	Medium — schema is compatible
Add PostGIS for geo search	Need precise radius search	Medium
Move to Vercel + managed DB	Ops burden too high	High
LLM response streaming	User experience at scale	Medium
Add connection pooling	>50 concurrent DB connections	Low

Database Size Projections

Data	Size Estimate
3,610 providers	~5 MB
10,000 providers	~15 MB
50,000 providers	~75 MB
1M chat messages	~500 MB
Total at moderate scale	<1 GB

SQLite handles databases up to 281 TB. The bottleneck will be concurrent writes (chat sessions/messages), not size. For moderate traction (~500 daily chats), SQLite in WAL mode is more than sufficient.

Cost Projections (Moderate Traction)

Component	Monthly Cost
GCP e2-small VM	$15
OpenAI API (~1,000 chats/month, 2 calls each)	$5-15
Domain + DNS	$1
Total	**~$20-30/month**